

Night shift

ar065 · 11 July 2026 · Draft

This is the **policy** the overnight agent obeys — separate from the plan (the science) and the log (the record). It runs unattended for at most one night, works the plan’s queue, and obeys the contract below.

What the night shift may do

- **Run only what is queued** by a human (queue entries with *origin: human*). Never invent an experiment, never run a *proposed* entry.
- **May propose.** It can draft new *proposed* queue entries for the morning review, but never runs them — proposing is a suggestion, not an action.
- **Publish nothing.** It ends by prepending a digest to the log and leaving its output for review. Nothing merges without a human — the morning gate decides.
- **Enforce its own budgets**, aborting a run at its wall-clock ceiling and ending the shift on any stop condition below.

Compute — RunPod is authorised for this program

Both local (Mac MPS, free) and RunPod are allowed. **This program explicitly authorises the night shift to launch RunPod pods and spend real money unattended** — a standing, deliberate exception to the default rule that cloud runs need per-run permission. It exists because the stability sweeps (exp061–063) are faster on a GPU and the scientist has accepted the cost.

The authorisation is **bounded, not open**: a hard *max cost per night* ceiling, a fixed GPU tier, and one pod at a time. When the ceiling is reached the shift stops, even mid-queue. Pods are reaped at shift end; a leaked pod is an anomaly for the morning gate.

The contract

```
budgets:
  epochs_per_run: 40          # ≈ one seed at native dt on MPS; less on RunPod
  wall_clock_per_run: 6h      # hard ceiling – kills a diverging run
  wall_clock_per_night: 8h
  seeds_default: 3           # one seed is an anecdote, not a result
scope:
  collection: spiking-heidelberg-digits # the only collection this shift may touch
  may_propose: true           # draft `proposed` entries; never run them
compute:
  local: allowed              # free, the default
  runpod: allowed             # EXPLICITLY authorised for this program – real spend
  gpu: "4090"
  max_cost_per_night_usd: 40  # hard $ ceiling; stop when reached
  max_pods_concurrent: 1
  reap_pods_at_end: true
stop_when:
  - queue_empty
  - night_budget_exhausted
  - cost_ceiling_reached
  - build_red_twice          # bail rather than thrash
```

The morning gate

Every night ends with a ruthless digest at the top of the log: what confirmed, what died by its own kill criterion, what anomaly (a leaked pod, an unexplained NaN, a blown budget) needs a

human. The review reads that first, and decides what — if anything — merges. The shift never merges its own work.